

Dilated pieces, Hecke correspondences and the second spectrum: the level-one lines of the pieces $\mathcal{P}_u$

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Abstract

Part IV of this series [5] showed that the pieces $\mathcal{P}_u(Y)$ of the off-diagonal of the pair singular series of an irreducible quadratic f oscillate at the spectral parameters of the even Maass cusp forms, and that the piece with dilation u lives at level u^2 . It left open “the size, in u , of the restricted geometric factor”. This note computes that factor for the level-one lines and for the newforms of level u . The piece with dilation u is the level-one seed of the smooth-window theorem, at length uY , summed over the $\Gamma_0(u^2)$ -invariant family $\mathcal{W}_u$ of forms of discriminant $4u^2D$ with $u^2 \mid a$ and $2u^2 \mid b$. The coefficient of the line t_j of a level-one form u_j in piece u is therefore a projection onto the oldform space $\{u_j(z), u_j(uz), u_j(u^2z)\}$, and for an odd prime u it equals the coefficient in piece 1 times an explicit rational function of $\lambda_j(u)$, $\sqrt{u}$ and $\chi_D(u)$: the Gram matrix of the oldforms (Iwaniec–Luo–Sarnak) and two local period identities for the Hecke neighbours of Heegner points (Gross–Kohnen–Zagier). The decay in u is set by the splitting type of u in $\mathbb{Q}(\sqrt{D})$ and not by the Hecke eigenvalue: the ratio is exactly $u^{-3/2}$ when u is inert, $-u^{-3/2}(1 - 2\lambda_j(u)u^{-1/2} + O(u^{-1}))$ when u splits, and $u^{-1/2}(1 - \lambda_j(u)u^{-1/2} + O(u^{-1}))$ when u is ramified; the ratio is multiplicative in squarefree u away from 2 and the primes whose square divides D , which contribute one joint factor, and at $u = 2$ the local constants depend on the class through the bilinear form modulo 2; so the main terms of the level-one lines are uniformly summable over u . The same formula with a newform v of level u gives its line in piece u , and the period that drives it vanishes when u is inert, because the Hecke and Fricke contributions cancel, and for every u when the Fricke sign is -1 . The piece splits into the $u = 1$ object at length uY , a part of level u and a part of genuinely level u^2 ; this explains the inert identity, and in the data each named line lies in the part the splitting assigns it. Both predictions carry no free parameter. On grids to $Y = 10^7$ all 21 pairs (u, D) tested for the first level-one line agree with the prediction within about one error estimate, the errors ranging from 0.01 to about 0.7 in the ratio, smallest at $u = 2$ and largest at $u = 5$, where the level- u^2 spectrum is dense. The first newform of level 3 agrees to a few percent in amplitude and a few hundredths of a radian in phase at the five discriminants where it is predicted, and is at the noise level at the two inert ones. The error term uniform in u , the open core of the conjecture of part III, is not touched here; part V [6] proves power savings for the pieces on three ranges of u and leaves a neighbourhood of $u = H^{1/3}$ open.

1 Introduction

The object. Notation as in part IV [5]. For a discriminant $D < 0$, a squarefree $u \geq 1$ and $Q_u(h) = u^2h^2 - D$, the unrestricted piece with dilation u and smooth window $w \in C_c^\infty((0, \infty))$

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is

$$S_u^w(Y) = \sum_{h \geq 1} w\left(\frac{h}{Y}\right) (\sigma_{-1}(Q_u(h)) - \mathbb{E}_u), \quad \sigma_{-1}(n) = \sum_{d|n} \frac{1}{d}, \quad \mathbb{E}_u = \sum_d \frac{\rho_u(d)}{d^2},$$

$\rho_u(d)$ the number of roots of $u^2 x^2 \equiv D \pmod{d}$. For $u = 1$ Theorem SMOOTH of part IV gives

$$\begin{aligned} \sqrt{Y} S_1^w(Y) &= |D|^{-1/4} \sum_j \text{Per}_D(u_j) \tilde{L}_j [A_j(Y) + \overline{A_j(Y)}] + \mathcal{E}^w(Y) + O(Y^{-2+\varepsilon}), \\ A_j(Y) &= \Gamma(-it_j) W_c\left(\frac{3}{2} + it_j\right) \left(\frac{\pi \sqrt{|D|}}{Y}\right)^{it_j}, \end{aligned} \tag{1}$$

the sum over an orthonormal basis of even Maass cusp forms of $\text{SL}_2(\mathbb{Z})$, with $\text{Per}_D(u_j) = \sum_Q u_j(z_Q)/|\Gamma_Q|$ over all classes of forms of discriminant $4D$, $\tilde{L}_j = \sum_k a_j(k) k^{-3/2}$, and W_c the Mellin factor of the window. Part IV, Section 7, showed that piece u lives at level u^2 and named the one missing ingredient: how the coefficients depend on u . Its footnote on $u = 3, 5$ recorded that the level-one lines are weak there and that the lines of $\Gamma_0(9)$ take over.

Results. Write $\lambda_j(u)$ for the Hecke eigenvalue ($T_u u_j = \lambda_j(u) u_j$, $a_j(1) = 1$) and $\chi = \chi_D(u)$ for the Kronecker symbol, so $\chi = 0$ iff $u \nmid D$.

Theorem 1 (the spectral formula for piece u). *Let $\mathcal{W}_u = \{[a, b, c]$ of discriminant $4u^2 D : u^2 \mid a, 2u^2 \mid b\}$ (all contents). Then $\mathcal{W}_u$ is $\Gamma_0(u^2)$ -invariant and, for every $\varepsilon > 0$,*

$$\sqrt{Y} S_u^w(Y) = \sqrt{u} |D|^{-1/4} \sum_v \text{Per}_{\mathcal{W}_u}(v) \tilde{L}_v [A_v(uY) + \overline{A_v(uY)}] + \mathcal{E}_u^w(Y) + O_{w,D,u,\varepsilon}(Y^{-2+\varepsilon}),$$

the sum over an orthonormal basis $\{v\}$ of the even Maass cusp forms of $\Gamma_0(u^2)$, with $\text{Per}_{\mathcal{W}_u}(v) = \sum_{Q \in \Gamma_0(u^2) \backslash \mathcal{W}_u} v(z_Q)/|\Gamma_0(u^2)_{z_Q}|$, $\tilde{L}_v$ formed with the Fourier coefficients of v at the cusp ∞ , and $\mathcal{E}_u^w$ the continuous spectrum, $\mathcal{E}_u^w(Y) \ll_{A,u} Y^{-1/2} (\log Y)^{-A}$. Every implied constant depends on u (through the spectral data of $\Gamma_0(u^2)$); nothing here is uniform in u , which is why the u -dependence of the coefficients is computed explicitly below.

Status: proved (Section 2): it is Theorem SMOOTH of part IV at level u^2 applied to an explicit identity between seeds; every constant checked numerically at $u = 2, 3, 5$.

Theorem 2 (the level-one lines). *Let u be an odd prime with $u^2 \nmid D$ and u_j an even Hecke-Maass form of level one. The coefficient of the line t_j in $\sqrt{Y} S_u^w(Y)$ equals the coefficient in $\sqrt{Y} S_1^w(Y)$ multiplied by u^{-it_j} and by*

$$\begin{aligned} r_j(u; D) &= \sqrt{u} P^\top (u(u+1) G)^{-1} L, \quad G = \begin{pmatrix} 1 & g & h \\ g & 1 & g \\ h & g & 1 \end{pmatrix}, \\ g &= \frac{\lambda_j(u) \sqrt{u}}{u+1}, \quad h = \frac{\lambda_j(u)^2 - 1 - 1/u}{u+1}, \quad L = (1, u^{-1}, u^{-2}), \quad P = (p_0, p_1, p_0), \\ p_1 &= \begin{cases} 2u & u \parallel D, \\ u - \chi & u \nmid D, \end{cases} \quad p_0 = \lambda_j(u) \sqrt{u} + \begin{cases} u - 1 & u \parallel D, \\ -(1 + \chi) & u \nmid D, \end{cases} \end{aligned} \tag{2}$$

where $u \parallel D$ means $u \mid D$, $u^2 \nmid D$. (For $u^2 \mid D$, and for $u = 2$ with every D , the entries of P depend on the class and are given by Proposition 8; the rest of (2) is unchanged.) Moreover

$$\begin{aligned} r_j(u; D) &= u^{-3/2} && (\text{inert}), \\ r_j(u; D) &= -u^{-3/2} (1 - 2\lambda_j(u) u^{-1/2} + O(u^{-1})) && (\text{split}), \\ r_j(u; D) &= u^{-1/2} (1 - \lambda_j(u) u^{-1/2} + (\lambda_j(u)^2 + 1) u^{-1} + O(u^{-3/2})) && (\text{ramified}), \end{aligned}$$

the implied constants absolute, and the inert identity exact for every $\lambda_j(u)$. The Hecke eigenvalue enters only the corrections; the decay is set by the splitting type.

Status: proved (Section 3); the asymptotics are read off (2) as a rational function of λ and $\sqrt{u}$. The inert identity has a one-line reason: for $\chi = -1$, $P = (\lambda\sqrt{u}, u+1, \lambda\sqrt{u}) = (u+1)(g, 1, g)$ is $(u+1)$ times the middle column of G , so $G^{-1}P = (u+1)e_2$ and $P^\top G^{-1}L = (u+1)/u$, whence $r = \sqrt{u}(u+1)/(u \cdot u(u+1)) = u^{-3/2}$. A reason that uses no Gram matrix is Remark 13: for inert u the level-one lines of piece u are exactly those of $\bar{w} S_1^w(uY)$ with $\bar{w} = 1/u$. The first version of this draft stated the asymptotic as $\lambda_j(u)/u$ for $u \nmid D$; that was wrong — the $\lambda_j(u)\sqrt{u}$ terms of p_0 and $p_1 L_2$ cancel exactly — and was caught by the author's reviewing conversation, and independently by a fresh model instance, both evaluating (2) at large u . The values of (2) agree with the direct spectral computation to four digits and with the data to the noise (Table 1).

Theorem 3 (the newform lines). *Let u be an odd prime and v an even Hecke–Maass newform of level u with Atkin–Lehner eigenvalue ε ($a_v(u) = -\varepsilon u^{-1/2}$). Its line in $\sqrt{Y} S_u^w(Y)$ is*

$$\sqrt{u} |D|^{-1/4} \left[\sum_{k=1}^2 \text{Per}_{\mathcal{W}_u}(v_k) \tilde{L}(v_k) \right] [A_v(uY) + \overline{A_v(uY)}],$$

$$A_v(uY) = \Gamma(-it_v) W_c\left(\frac{3}{2} + it_v\right) \left(\frac{\pi \sqrt{|D|}}{uY} \right)^{it_v},$$

i.e. an amplitude $2\sqrt{u}|D|^{-1/4} |\sum_k \text{Per}(v_k) \tilde{L}(v_k)| |\Gamma(-it_v) W_c(\frac{3}{2} + it_v)|$ and a phase shifted by $-t_v \log u$; $\{v_1, v_2\}$ the orthonormalisation of $\{v(z), v(uz)\}$ on $\Gamma_0(u^2) \backslash \mathbb{H}$, whose Gram matrix has off-diagonal entry $a_v(u)\sqrt{u}/u = -\varepsilon/u$ relative to $\langle v, v \rangle$. If u is inert in $\mathbb{Q}(\sqrt{D})$, or if $\varepsilon = -1$, then $\text{Per}_{\mathcal{W}_u}(v) = \text{Per}_{\mathcal{W}_u}(v(u \cdot)) = 0$ and the line is absent.

Status: proved: the formula is Theorem 1 restricted to the two-dimensional oldspace, the Gram entry is Lemma 10(iii), and the vanishing is Lemma 12. Tested on seven discriminants (Table 2): the period vanishes identically for inert u in the computation, and in the data the line is at the noise level there. At $u = 5$ the two even newforms with $\varepsilon = -1$ below $R = 8.6$ are absent from the data and the three with the largest lines have $\varepsilon = +1$ (Section 4).

Corollary 4 (uniform summability of the level-one main terms). *In Theorem A' of part III the piece with dilation u enters at length H/u , so its level-one main terms are of size $(H/u)^{1/2} |r_j(u; D)|$ times the $u = 1$ amplitude. For a fixed level-one form u_j and bounded weights $w(u)$, $\sum_u |w(u)| (H/u)^{1/2} |r_j(u; D)| \ll_{D, \varepsilon} H^{1/2}$ over all squarefree u . By Proposition 11, $|r_j(u; D)| = |r_j(u_e; D)| \prod_{p|u/u_e} |r_j(p; D)|$, where u_e divides the product of 2 and the primes whose square divides D , so $|r_j(u_e; D)|$ takes finitely many values; they can exceed 1 (Remark 9). By Theorem 2 the product is $\ll_{D, \varepsilon} u^{-3/2+\varepsilon} \prod_{p|(u, D)} p$, and $\sum_u u^{-2+\varepsilon}$ converges. (The weights of part III are $w(u) = \lambda(u)\omega(u) \ll u^\varepsilon$; the conclusion is unchanged.)*

Remark 5 (what this does and does not settle). Part III's conjecture needs the pieces with $u \leq H^{1/2+\varepsilon}$ to sum to $o(H \log H)$ with their errors. The corollary says the level-one main terms do so with room to spare; the newform lines of level u are new at every u (Theorem 3 gives their size for each u but not a bound uniform in the newform), the cycloidal-group forms of level u^2 [7] are not covered at all, and the error term of Remark GAPS of part IV is untouched here (part V [6] bounds it on three ranges of u). What has changed is that the main-term side of the u -problem is Hecke theory on Heegner divisors, and it decays.

Method and checks. Section 2 proves Theorem 1. Section 3 states the three lemmas behind Theorem 2 with what is proved of each. Section 4 reports the tests: the ratio (2) against a direct computation of the oldform projection with the Maass forms' values (no closed form used), and

both against the pieces computed to $Y = 10^7$; then the newform of level 3; then the case $u = 2$ and the composite $u = 6$; then the components of the piece and the precision of the data. Section 5 lists what is open. Scripts and data are in the repository (`research/explore/`, `hecke-oldform-predict.py`, `hecke-classwise.py`, `hecke-newform-u3.py`, `hecke-ratio-sharp.py`, `hecke-components.ts`, `newform-fricke-check.py`).

2 The seed identity and the family $\mathcal{W}_u$

The pairs (d, x) with $u^2 x^2 \equiv D \pmod{d}$ are the forms $Q = [d, 2ux, (u^2 x^2 - D)/d]$ of discriminant $4D$ with $2u \mid b$; write $\mathcal{H}^{(u)}$ for this sub-family of the set $\mathcal{H}$ of all forms of discriminant $4D$, and $z_Q = (-ux + i\sqrt{|D|})/d$. Poisson summation in h , exactly as in the proof of Theorem SMOOTH, writes $S_u^w(Y)$ as $\sum_{(d,x)} \frac{Y}{d^2} \sum_{k \neq 0} \hat{w}(kY/d) e(kx/d)$, and in the coordinates of $z = z_Q$ ($y = \sqrt{|D|}/d$, $x/d = -\Re z/u$) the summand is

$$\frac{Yy^2}{|D|} \sum_{k \neq 0} \hat{w}\left(\frac{kYy}{\sqrt{|D|}}\right) e\left(-\frac{k\Re z}{u}\right) = u \Psi_{uY}^w(z/u),$$

$\Psi_{Y'}^w(\tau) = \frac{Y'(\Im \tau)^2}{|D|} \sum_{k \neq 0} \hat{w}(kY'\Im \tau/\sqrt{|D|}) e(-k\Re \tau)$ the level-one seed of part IV at length Y' . The point $\tau = z_Q/u = (-2u^2x + i\sqrt{4u^2|D|})/(2u^2d)$ is the Heegner point of the form $Q' = [u^2d, 2u^2x, (u^2x^2 - D)/d]$ of discriminant $4u^2D$, and $Q \mapsto Q'$ is a bijection of $\mathcal{H}^{(u)}$ onto $\mathcal{W}_u$. For $\gamma \in \Gamma_0(u^2)$ the middle coefficient transforms as $b' = 2a\alpha\beta + b(\alpha\delta + \beta\gamma) + 2c\gamma\delta \equiv b\alpha\delta \equiv b \pmod{2u^2}$, since $u^2 \mid a$, $u^2 \mid \gamma$ and $\alpha\delta \equiv 1 \pmod{u^2}$; the condition $u^2 \mid a$ is $\Gamma_0(u^2)$ -stable by Lemma SUBFAMILY of part IV. So $\mathcal{W}_u$ is a finite union of $\Gamma_0(u^2)$ -orbits, the translation $\tau \mapsto \tau + 1$ is $x \mapsto x - d$, and $S_u^w(Y) = u \sum_{\tau \in \mathcal{W}_u/\mathbb{Z}} \Psi_{uY}^w(\tau)$. The proof of Theorem SMOOTH applies verbatim on $\Gamma_0(u^2)\backslash\mathbb{H}$ (its cusp at ∞ has width one, so the pairing with the Fourier expansions at ∞ is unchanged), with Y replaced by uY ; the factor u and $(uY)^{-1/2}$ combine to $\sqrt{u}Y^{-1/2}$, and $(\pi\sqrt{|D|}/(uY))^{it_v}$ carries the phase $-t_v \log u$. This proves Theorem 1. For $f_m(z) = u_j(u^m z)$ the coefficient at index $u^m n$ is $u^{m/2} a_j(n)$, so $\tilde{L}_{f_m} = u^{-m} \tilde{L}_j$, which is the vector L of Theorem 2.

3 The three lemmas

Throughout u is an odd prime and $\text{Per}_D(f) = \sum_{Q \in \mathcal{H}/\text{SL}_2(\mathbb{Z})} f(z_Q)/|\Gamma_Q|$.

Lemma 6 (multiplicity). *Let u be an odd prime with $u^2 \nmid D$ (for odd u , $u \parallel D$ already forces u prime to the conductor of the order of discriminant $4D$). For every $\text{SL}_2(\mathbb{Z})$ -class C of forms of discriminant $4D$, $\sum_{Q' \in \Gamma_0(u^2)\backslash\mathcal{W}_u, u z_{Q'} \in C} |\Gamma_0(u^2)_{z_{Q'}}|^{-1} = m_u(D) |\Gamma_C|^{-1}$ with $m_u(D) = 2u$ if $u \parallel D$ and $m_u(D) = u - \chi_D(u)$ if $u \nmid D$. Equivalently $\text{Per}_{\mathcal{W}_u}(u_j(u \cdot)) = m_u(D) \text{Per}_D(u_j)$. Stabilisers are taken in PSL_2 throughout (sizes 1, 2, 3); since $-I$ lies in every group considered, the convention only has to be the same on both sides of each identity.*

Proof. Write $(Q \circ \gamma)(v) = Q(\gamma v)$ (a right action), $C = Q_0 \circ \Gamma$ with $\Gamma = \text{SL}_2(\mathbb{Z})$, $S = \{\gamma : Q_0 \circ \gamma = Q_0\}$, and let B be the bilinear form of Q_0 , $B(v, w) = av_1w_1 + \frac{b}{2}(v_1w_2 + v_2w_1) + cv_2w_2$, of discriminant D .

The transported group. With $\alpha = \begin{pmatrix} 1 & 0 \\ 0 & u \end{pmatrix}$, $z \mapsto z/u$, one has $\alpha^{-1}\Gamma_0(u^2)\alpha = G_u := \Gamma_0(u) \cap \Gamma^0(u) = \{u \mid b, u \mid c\}$, of index $u(u+1)$. The bijection $Q \mapsto Q' = [u^2a, ub, c]$ of $\mathcal{H}^{(u)}$ onto $\mathcal{W}_u$ has $z_{Q'} = z_Q/u$ and intertwines the action of G_u on $\mathcal{H}^{(u)}$ with that of $\Gamma_0(u^2)$ on $\mathcal{W}_u$, orbits and stabilisers included; and $\mathcal{H}^{(u)}$ is G_u -stable, since $b(Q \circ g) \equiv b\alpha\delta \equiv b \pmod{u}$ for $g \in G_u$. So we count G_u -orbits in $C \cap \mathcal{H}^{(u)}$.

Cosets are ordered pairs of lines. $\Gamma \rightarrow \mathrm{SL}_2(\mathbb{F}_u)$ is onto and G_u is the preimage of the diagonal torus T , so $\Gamma/G_u \cong \mathrm{SL}_2(\mathbb{F}_u)/T$: a matrix with columns (v, w) modulo $(v, w) \mapsto (\lambda v, \lambda^{-1} w)$ is the ordered pair of distinct lines $(\ell_1, \ell_2) = (\langle v \rangle, \langle w \rangle)$ in $\mathbb{P}^1(\mathbb{F}_u)$, $u(u+1)$ pairs in all.

The family condition is orthogonality. The middle coefficient of $Q_0 \circ \gamma$ is $2B(\gamma e_1, \gamma e_2)$; so $Q_0 \circ \gamma \in \mathcal{H}^{(u)}$ iff $\ell_1 \perp_B \ell_2$, a condition on the right coset γG_u and unchanged under $\gamma \mapsto s\gamma$, $s \in S$.

The mass formula. $C \cong S \backslash \Gamma$, the G_u -orbits in C are the double cosets $S\gamma G_u$, with stabiliser $G_u \cap \gamma^{-1} S \gamma$, and each double coset consists of $|S|/|G_u \cap \gamma^{-1} S \gamma|$ right cosets $\gamma' G_u$. Hence for every function φ on Γ that is left- S - and right- G_u -invariant,

$$\sum_{G_u\text{-orbits } O \subset C} \frac{\varphi(O)}{|\mathrm{Stab}_{G_u}(O)|} = \frac{1}{|S|} \sum_{\gamma \in \Gamma/G_u} \varphi(\gamma). \quad (3)$$

With $\varphi = \mathbf{1}[\ell_1 \perp \ell_2]$ this gives $m_u(D) = \#\{(\ell_1, \ell_2) \text{ distinct} : \ell_1 \perp_B \ell_2\}$. If $u \nmid D$, B is nondegenerate modulo u ; every ℓ_1 has a unique $\ell_1^\perp$, admissible iff $\ell_1^\perp \neq \ell_1$, i.e. iff ℓ_1 is not isotropic; the isotropic lines are the roots of Q_0 modulo u , $1 + \chi_D(u)$ of them; so $m = u+1 - (1 + \chi) = u - \chi$. If $u \parallel D$, then $Q_0 \not\equiv 0 \pmod{u}$ for every class (a content divisible by u would force $u^2 \mid D$) and B has rank one modulo u with radical line r (the double root); $B(v, w) = 0$ iff $\langle v \rangle = r$ or $\langle w \rangle = r$, giving $u + u = 2u$ ordered pairs. $\square$

Lemma 7 (the period of the dilated family). *Under the hypotheses of Lemma 6, $\mathrm{Per}_{\mathcal{W}_u}(u_j) = \mathrm{Per}_{\mathcal{W}_u}(u_j(u^2 \cdot)) = (\lambda_j(u)\sqrt{u} + c_u(D))\mathrm{Per}_D(u_j)$ with $c_u(D) = u - 1$ if $u \parallel D$ and $c_u(D) = -(1 + \chi_D(u))$ if $u \nmid D$.*

Proof. In the notation of the previous proof, $z_{Q_0 \circ \gamma} = \gamma^{-1} z_0$, so $z_{Q_0 \circ \gamma}/u = \alpha \gamma^{-1} z_0$ with $\alpha \gamma^{-1}$ of determinant u . Since $\alpha M \alpha^{-1} = \begin{pmatrix} a & b/u \\ uc & d \end{pmatrix}$ is integral iff $u \mid b$, the coset $\Gamma \alpha \gamma^{-1}$ depends on γ only modulo right multiplication by $\Gamma^0(u)$, which fixes the second column line $\ell_2 = \langle \gamma e_2 \rangle$ and moves the first; the map $\gamma \mapsto \Gamma \alpha \gamma^{-1}$ is a bijection of $\Gamma/\Gamma^0(u)$ onto the $u+1$ cosets $\Gamma \backslash \Delta_u$ of the Hecke operator. (Check: $\gamma = \begin{pmatrix} 1 & -k \\ 0 & 1 \end{pmatrix}$ has $\ell_1 = \langle e_1 \rangle$ fixed, $\ell_2 = \langle (-k, 1) \rangle$, and $\alpha \gamma^{-1} z = (z + k)/u$, u distinct neighbours.) Hence $u_j(\alpha \gamma^{-1} z_0) =: u_j(h_{\ell_2}(z_0))$ depends only on ℓ_2 , the points $h_\ell(z_0)$, $\ell \in \mathbb{P}^1(\mathbb{F}_u)$, are the $u+1$ Hecke neighbours of z_0 , and $\sum_\ell u_j(h_\ell(z_0)) = \sqrt{u} \lambda_j(u) u_j(z_0)$. Applying (3) with $\varphi(\gamma) = u_j(\alpha \gamma^{-1} z_0) \mathbf{1}[\ell_1 \perp \ell_2]$ and summing over the classes,

$$\mathrm{Per}_{\mathcal{W}_u}(u_j) = \sum_C \frac{1}{|S_C|} \sum_{\ell_2} N(\ell_2) u_j(h_{\ell_2}(z_C)), \quad N(\ell_2) = \#\{\ell_1 \neq \ell_2 : \ell_1 \perp \ell_2\}.$$

If $u \nmid D$ then $N(\ell_2) = 1$ for non-isotropic and 0 for isotropic ℓ_2 , so the inner sum is $\sqrt{u} \lambda_j(u) u_j(z_C) - \sum_{\ell \text{ isotropic}} u_j(h_\ell(z_C))$. The isotropic lines are the eigenlines of the order $\mathcal{O}_{4D}$ acting on L_C/uL_C , $L_C = \mathbb{Z} + \mathbb{Z}z_C$, i.e. the $1 + \chi$ sublattices $\mathfrak{p}L_C$ for the primes $\mathfrak{p} \mid u$ of the order; their Heegner points have discriminant $4D$ again, with class $[\mathfrak{p}] \cdot C$ [1, II]. Since $C \mapsto \mathfrak{p}C$ permutes the classes of each order and preserves $|S_C|$ (the units of the order), $\sum_C |S_C|^{-1} \sum_{\text{iso}} u_j(h_\ell(z_C)) = (1 + \chi) \mathrm{Per}_D(u_j)$, and the claim follows. If $u \parallel D$, $N(r) = u$ and $N(\ell_2) = 1$ for $\ell_2 \neq r$, so the inner sum is $\sqrt{u} \lambda_j(u) u_j(z_C) + (u-1) u_j(h_r(z_C))$, and $h_r(z_C) = z_{\mathfrak{p}C}$ for the ramified prime $\mathfrak{p}$, again a permutation of the classes: $\mathrm{Per}_{\mathcal{W}_u}(u_j) = (\sqrt{u} \lambda_j(u) + u-1) \mathrm{Per}_D(u_j)$. Finally $u^2 z_{Q'} = u z_Q = \alpha' \gamma^{-1} z_0$ with $\alpha' = \begin{pmatrix} u & 0 \\ 0 & 1 \end{pmatrix}$, whose Hecke coset is determined by ℓ_1 ; the counts $\#\{\ell_2 \neq \ell_1 : \ell_1 \perp \ell_2\}$ equal the $N(\ell_2)$ by the symmetry of B , which gives $\mathrm{Per}_{\mathcal{W}_u}(u_j(u^2 \cdot)) = \mathrm{Per}_{\mathcal{W}_u}(u_j)$. (The same equality follows from the Fricke involution $\tau \mapsto -1/(u^2 \tau)$ of $\Gamma_0(u^2)$, which maps $[u^2 a', 2u^2 b', c]$ to $[u^2 c, -2u^2 b', a']$ — the pairing of the divisor d with $Q_u(x)/d$ — and so permutes $\mathcal{W}_u$.) $\square$

Proposition 8 (the class-by-class form of Lemmas 6 and 7). *Let u be any prime and $D < 0$ any discriminant (no hypothesis on u and D). For a class C of forms of discriminant $4D$ with representative $Q_C = [a, b, c]$ and bilinear form B_C , let*

$$m(C) = \#\{(\ell_1, \ell_2) \text{ distinct} : B_C(\ell_1, \ell_2) \equiv 0 \pmod{u}\},$$

$$N_C(\ell_2) = \#\{\ell_1 \neq \ell_2 : B_C(\ell_1, \ell_2) \equiv 0 \pmod{u}\},$$

over the lines of $\mathbb{P}^1(\mathbb{F}_u)$, and let h_ℓ be the Hecke neighbour fixed by ℓ : $h_{\langle(k,1)\rangle}z = (z - k)/u$, $h_{\langle(1,0)\rangle}z = -1/(uz)$. Then

$$\text{Per}_{\mathcal{W}_u}(u_j(u \cdot)) = \sum_C \frac{m(C)}{|S_C|} u_j(z_C),$$

$$\text{Per}_{\mathcal{W}_u}(u_j) = \text{Per}_{\mathcal{W}_u}(u_j(u^2 \cdot)) = \sum_C \frac{1}{|S_C|} \sum_\ell N_C(\ell) u_j(h_\ell z_C),$$

and Theorem 2 holds with $p_1 = \text{Per}_{\mathcal{W}_u}(u_j(u \cdot))/\text{Per}_D(u_j)$, $p_0 = \text{Per}_{\mathcal{W}_u}(u_j)/\text{Per}_D(u_j)$. For $u = 2$ the counts depend only on $(a, b/2, c)$ modulo 2: with $b' = b/2$,

$$m(C) = 2([b' \equiv 0] + [a \equiv b'] + [c \equiv b']), \quad N_C(\langle e_2 \rangle) = [b' \equiv 0] + [c \equiv b'],$$

$$N_C(\langle e_1 + e_2 \rangle) = [a \equiv b'] + [c \equiv b'], \quad N_C(\langle e_1 \rangle) = [b' \equiv 0] + [a \equiv b'];$$

so $m(C) = 4$ for a primitive class of even D , 6 for a class of even content, 2 for a class of odd D that is not of even content, and 0 for a class of even content and odd D (where B_C is alternating modulo 2).

Proof. Steps 1–4 of the proof of Lemma 6 and the first half of the proof of Lemma 7 (the coset bijection, the orthogonality condition, the mass formula (3) and the identification of $\alpha\gamma^{-1}z_0$ with the neighbour $h_{\ell_2}z_0$) use neither the parity of u nor any hypothesis on D ; the two displays are (3) with $\varphi = \mathbf{1}[\ell_1 \perp \ell_2]$ and $\varphi = u_j(\alpha\gamma^{-1}z_0)\mathbf{1}[\ell_1 \perp \ell_2]$, and the equality of the periods of f_0 and f_2 is the symmetry of B_C as before. Only the evaluation of the counts used the hypotheses: for odd u with $u^2 \nmid D$ the reduction $B_C \bmod u$ is determined by D up to equivalence, hence the counts are class-independent; for $u = 2$ the three lines are $\langle e_1 \rangle, \langle e_2 \rangle, \langle e_1 + e_2 \rangle$ with $B_C(e_1, e_2) = b'$, $B_C(e_1, e_1 + e_2) = a + b'$, $B_C(e_2, e_1 + e_2) = b' + c$, which gives the table (a nondegenerate symmetric form over $\mathbb{F}_2$ is not determined by its determinant: $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ are inequivalent, which is why the counts depend on the class); for $u^2 \mid D$ the classes of content divisible by u have $B_C \equiv 0$ and $m(C) = u(u + 1)$. The case distinction for $m(C)$ at $u = 2$ is read off $b'^2 - ac = D$: for even D , either b' is odd and a, c are odd, or b' is even and exactly one of a, c is even, both giving $m = 4$, or a, b, c are all even ($m = 6$); for odd D , either b' is even and a, c odd, or b' odd and exactly one of a, c even, both giving $m = 2$, or b' is odd and a, c even, the even-content classes, where $B_C \bmod 2$ is alternating and $m = 0$. $\square$

Remark 9. The two displays of Proposition 8 were checked against the direct enumeration of the $\Gamma_0(u^2)$ -orbits of $\mathcal{W}_u$ (`hecke-classwise.py`) for $u = 2$ and nine discriminants $-3 \geq D \geq -20$, and for $u = 3$ with $D = -9, -27, -36$; the two routes agree to the four digits printed. At $u = 2$ the resulting p_1 takes the values 4.913 ($D \equiv 0 \pmod{4}$), 5.278 ($D/4 \equiv 1 \pmod{4}$), 5.565 ($D/4 \equiv 0 \pmod{4}$), 0.321 ($D \equiv 1 \pmod{8}$), 1.373 ($D \equiv 5 \pmod{8}$): by the table, $p_1 = 4 + 2\text{Per}_D^{(2)}/\text{Per}_D$ for even D and $2 - 2\text{Per}_D^{(2)}/\text{Per}_D$ for odd D , where $\text{Per}_D^{(2)}$ is the period over the classes of even content, i.e. over the forms $2Q'$ with Q' of discriminant D ; the observed values give $\text{Per}_D/\text{Per}_D^{(2)} = \sqrt{2}\lambda_j(2) - \chi_D(2)$ to four digits when D is odd or $D/4 \equiv 2, 3 \pmod{4}$ (when $D/4$ is not itself a discriminant), as the Hecke relation at 2 between the periods of discriminants

D and $4D$ suggests, and a different value when $D/4$ is a discriminant (1.564 at $D = -12$, 1.278 at $D = -16$), where that relation has a further term. We have not used this. For $u = 3$ and $9 \mid D$ the ratio $r_1(3; D)$ exceeds 1 (4.22 at $D = -27$, 1.75 at $D = -9, -36$) because the classes of content 3 have $m(C) = 12$.

Lemma 10 (the Gram matrix). *On $\Gamma_0(u^2) \backslash \mathbb{H}$, with $f_m(z) = u_j(u^m z)$: (i) $\langle f_0, f_0 \rangle = u(u+1) \|u_j\|_{\text{SL}_2(\mathbb{Z})}^2$ and $\langle f_1, f_0 \rangle = \langle f_2, f_1 \rangle = \frac{\lambda_j(u)\sqrt{u}}{u+1} \langle f_0, f_0 \rangle$; (ii) $\langle f_2, f_0 \rangle = \frac{\lambda_j(u)^2 - 1 - 1/u}{u+1} \langle f_0, f_0 \rangle$; (iii) for a newform v of level u with Fricke eigenvalue ε , $\langle v(u \cdot), v \rangle = -\frac{\varepsilon}{u} \langle v, v \rangle$.*

Proof. Write $\text{Tr}_H^G F = \sum_{\gamma \in H \backslash G} F \circ \gamma$ for $H \subset G$ of finite index, so that $\langle F, \Phi \rangle_H = \langle \text{Tr}_H^G F, \Phi \rangle_G$ for G -invariant Φ , and $\langle \Phi, \Psi \rangle_H = [G : H] \langle \Phi, \Psi \rangle_G$ for G -invariant Φ, Ψ . (i) The cosets $\Gamma_0(u) \backslash \Gamma$ are $\begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix}$, $b \bmod u$, and S , and the points $u\gamma z$ over them are the $u+1$ Hecke neighbours of z ; so $\text{Tr}_{\Gamma_0(u)}^\Gamma (u_j(u \cdot)) = \sqrt{u} \lambda_j(u) u_j$, $\langle u_j(uz), u_j \rangle_{\Gamma_0(u)} = \sqrt{u} \lambda_j(u) \|u_j\|_\Gamma^2 = \frac{\sqrt{u} \lambda_j(u)}{u+1} \|u_j\|_{\Gamma_0(u)}^2$ [3, Lemma 2.4], and the ratio persists on $\Gamma_0(u^2)$ since both functions are $\Gamma_0(u)$ -invariant; $\langle f_2, f_1 \rangle$ reduces to the same by $z \mapsto z/u$, which carries $\Gamma_0(u^2) \backslash \mathbb{H}$ to $(\Gamma_0(u) \cap \Gamma^0(u)) \backslash \mathbb{H}$ of the same index. (ii) Trace from $\Gamma_0(u^2)$ to $\Gamma_0(u)$ over $\gamma_b = \begin{pmatrix} 1 & 0 \\ ub & 1 \end{pmatrix}$: $f_2(\gamma_b z) = u_j(\begin{pmatrix} u & 0 \\ b & 1 \end{pmatrix} uz)$, and $\begin{pmatrix} u & 0 \\ b & 1 \end{pmatrix} \in \Gamma \begin{pmatrix} 1 & k \\ 0 & u \end{pmatrix}$ with $k \equiv b^{-1}$ for $b \neq 0$, so $f_2(\gamma_b z) = u_j((uz + k)/u)$ for $b \neq 0$ and $u_j(u^2 z)$ for $b = 0$; by the Hecke relation at the point uz , $\text{Tr} f_2 = \sum_{k \bmod u} u_j((uz + k)/u) - u_j(z) + u_j(u^2 z) = \sqrt{u} \lambda_j(u) u_j(uz) - u_j(z)$. Hence $\langle f_2, f_0 \rangle_{\Gamma_0(u^2)} = \sqrt{u} \lambda_j(u) \langle f_1, f_0 \rangle_{\Gamma_0(u)} - \|u_j\|_{\Gamma_0(u)}^2 = (\frac{\lambda_j(u)^2 u}{u+1} - 1) \|u_j\|_{\Gamma_0(u)}^2 = \frac{\lambda_j(u)^2 - 1 - 1/u}{u+1} \langle f_0, f_0 \rangle$. (iii) The same trace gives $\text{Tr} v(u \cdot) = \sum_b v(\begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix} uz)$, and $\sum_b v(\begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix} w) = -v(-1/w)$ because a newform has vanishing trace to level one; with $w = uz$, $\text{Tr} v(u \cdot) = -v(-1/(uz)) = -\varepsilon v(z)$, so $\langle v(uz), v \rangle_{\Gamma_0(u^2)} = -\varepsilon \|v\|_{\Gamma_0(u)}^2 = -\frac{\varepsilon}{u} \|v\|_{\Gamma_0(u^2)}^2$. $\square$

Proposition 11 (multiplicativity in u). *Let u be squarefree and $D < 0$. Call a prime $p \mid u$ exceptional if $p = 2$ or $p^2 \mid D$, and let u_e be the product of the exceptional primes dividing u . Write $\{f_d(z) = u_j(dz) : d \mid u^2\}$ for the oldform space of u_j at level u^2 , $d = \prod_p p^{a_p}$ with $a_p \in \{0, 1, 2\}$, and $\mathbf{P}_d = \text{Per}_{\mathcal{W}_u}(f_d)$, $\mathbf{G}_{dd'} = \langle f_d, f_{d'} \rangle_{\Gamma_0(u^2)}$, $\mathbf{L}_d = \tilde{L}_{f_d} = \tilde{L}_j/d$. Then*

$$\frac{\mathbf{P}_d}{\text{Per}_D(u_j)} = \frac{\text{Per}_{\mathcal{W}_{u_e}}(f_{d_e})}{\text{Per}_D(u_j)} \prod_{p \mid u/u_e} p_{d_p}^{(p)},$$

$$\frac{\mathbf{G}_{dd'}}{\langle f_1, f_1 \rangle} = \prod_{p \mid u} \hat{G}_{a_p a'_p}^{(p)}, \quad \langle f_1, f_1 \rangle = \prod_{p \mid u} p(p+1) \|u_j\|^2,$$

where $d_e = \prod_{p \mid u_e} p^{a_p}$, $(p_0^{(p)}, p_1^{(p)}, p_2^{(p)}) = (p_0, p_1, p_0)$ is the vector P of Theorem 2 at the non-exceptional prime p , and $\hat{G}^{(p)}$ is the normalised 3×3 Gram matrix of Lemma 10 at p . Consequently $r_j(u; D) = r_j(u_e; D) \prod_{p \mid u/u_e} r_j(p; D)$. In particular $r_j(u; D) = \prod_{p \mid u} r_j(p; D)$ when u has at most one exceptional prime, whose factor is given by Proposition 8.

Proof. The periods. Steps 1–4 of the proof of Lemma 6 hold verbatim for squarefree u : $G_u = \Gamma_0(u) \cap \Gamma^0(u)$ is the preimage of $\prod_{p \mid u} T_p$ under $\Gamma \rightarrow \prod_p \text{SL}_2(\mathbb{F}_p)$, so a coset γG_u is a tuple $(\ell_1^{(p)}, \ell_2^{(p)})_{p \mid u}$ of ordered pairs of distinct lines, and the family condition $u \mid B(\gamma e_1, \gamma e_2)$ is $\ell_1^{(p)} \perp \ell_2^{(p)}$ for every p . For $d \mid u^2$ the point $d \cdot z_{Q_0} \gamma / u$ is $\text{diag}(d, u) \gamma^{-1} z_0$, and the p -part of $\text{diag}(d, u)$ is a scalar if $a_p = 1$, $\text{diag}(1, p)$ if $a_p = 0$ and $\text{diag}(p, 1)$ if $a_p = 2$; so the Γ -coset of $\text{diag}(d, u) \gamma^{-1}$ is determined by the lines $\ell_2^{(p)}$ ($a_p = 0$) and $\ell_1^{(p)}$ ($a_p = 2$) alone, and the point is the composite Hecke neighbour $\prod_{a_p \neq 1} h^{(p)} z_0$, the Heegner point of the sublattice of L_C cut out by those lines (the $h^{(p)}$ commute: sublattices of coprime index intersect). The mass formula (3) thus gives

$$\text{Per}_{\mathcal{W}_u}(f_d) = \sum_C \frac{1}{|S_C|} \prod_{a_p=1} m_p(C) \sum_{(\ell^{(p)})_{a_p \neq 1}} \prod_{a_p \neq 1} N_C^{(p)}(\ell^{(p)}) u_j \left(\prod_{a_p \neq 1} h_{\ell^{(p)}}^{(p)} z_C \right),$$

with $m_p, N_C^{(p)}$ as in Proposition 8. For a non-exceptional $p \mid u$ the inner operator is, as in the proof of Lemma 7, $\sqrt{p}\lambda_j(p) - \text{Iso}_p$ ($p \nmid D$) or $\sqrt{p}\lambda_j(p) + (p-1)\text{Ram}_p$ ($p \parallel D$), where Iso_p and Ram_p replace z_C by $z_{\mathfrak{p}C}$ for the primes $\mathfrak{p} \mid p$ of the order. The operator $\sqrt{p}\lambda_j(p)$ (the Hecke operator T_p) commutes with everything; and $C \mapsto \mathfrak{p}C$ is a permutation of the classes preserving $|S_C|$ under which every other prime's operator is transported: $(L_C/qL_C, B_C) \cong (\mathfrak{p}L_C/q\mathfrak{p}L_C, B_{\mathfrak{p}C})$ for $q \neq p$, since $[L_C : \mathfrak{p}L_C] = p$ is prime to q and $Q_{\mathfrak{p}C} = Q_C/p$ with p a unit modulo q ; under this isomorphism the index- q sublattice $M \subset L_C$ corresponds to $\mathfrak{p}M = M \cap \mathfrak{p}L_C$, so $N_{\mathfrak{p}C}^{(q)}(\mathfrak{p}M) = N_C^{(q)}(M)$ and $h_M^{(q)}h_{\mathfrak{p}}^{(p)}z_C = z_{\mathfrak{p}M} = h_{\mathfrak{p}M}^{(q)}z_{\mathfrak{p}C}$. Summing over C one prime at a time therefore extracts the factor $p_{a_p}^{(p)}$ for each non-exceptional p . The transport uses only that p is a unit modulo the other primes, not that their operators are class-independent, so what is left is $\sum_C |S_C|^{-1} \prod_{q|u_e} \text{Op}_{C,a_q}^{(q)} u_j(z_C)$, which is $\text{Per}_{\mathcal{W}_{u_e}}(f_{d_e})$ by the same mass formula at level u_e^2 . An exceptional prime cannot be extracted this way, because its counts depend on the class and its operator is not a Hecke operator plus a permutation of the classes; so the exceptional primes stay together.

The Gram matrix. For $a, b \geq 1$ put $g = (a, b)$, $a' = a/g$, $b' = b/g$. Unfolding against the Eisenstein series of $\Gamma_0(u^2)$ at ∞ (Rankin–Selberg), $\langle u_j(a \cdot), u_j(b \cdot) \rangle_{\Gamma_0(u^2)}$ is a constant depending only on u_j and u^2 times the residue at $s = 1$ of $\sum_{a|m, b|m} \sqrt{ab} \lambda_j(m/a) \lambda_j(m/b) m^{-s} = (a'b')^{-1/2} (ga'b')^{1-s} \sum_k \lambda_j(a'k) \lambda_j(b'k) k^{-s}$. Since $k \mapsto \lambda_j(a'k) \lambda_j(b'k)$ is multiplicative and its Euler factors at $p \nmid a'b'$ are those of $\sum \lambda_j(k)^2 k^{-s}$, the ratio of residues is

$$\frac{\langle u_j(a \cdot), u_j(b \cdot) \rangle}{\langle u_j, u_j \rangle_{\Gamma_0(u^2)}} = (a'b')^{-1/2} \prod_{p|a'b'} R_p(v_p(a'b')), \quad R_p(k) = \frac{\sum_{e \geq 0} \lambda_j(p^{k+e}) \lambda_j(p^e) p^{-e}}{\sum_{e \geq 0} \lambda_j(p^e)^2 p^{-e}},$$

the local series converging absolutely by the bound $|\lambda_j(p^e)| \leq (e+1)p^{7e/64}$. With $a = d$, $b = d'$ this is $\prod_p p^{-|a_p - a'_p|/2} R_p(|a_p - a'_p|)$, a tensor product of Toeplitz 3×3 matrices, whose entries must be those of Lemma 10 (take $u = p$): $p^{-1/2} R_p(1) = \lambda_j(p) \sqrt{p}/(p+1)$ and $p^{-1} R_p(2) = (\lambda_j(p)^2 - 1 - 1/p)/(p+1)$. (Checked numerically from the Hecke relations at $p = 2, 3, 5$; `hecke-classwise.py`.) The normalisation is $[\Gamma : \Gamma_0(u^2)] = \prod p(p+1)$.

The ratio. $\mathbf{L} = \bigotimes_p (1, p^{-1}, p^{-2}) \tilde{L}_j$, so $\sqrt{u} \mathbf{P}^\top \mathbf{G}^{-1} \mathbf{L}$ factors into the same expression at level u_e^2 times $\prod_{p|u/u_e} \sqrt{p} P^{(p)\top} (p(p+1) \hat{G}^{(p)})^{-1} L^{(p)}$, and dividing by the $u = 1$ coefficient $\text{Per}_D(u_j) \tilde{L}_j / \|u_j\|^2$ gives $r_j(u_e; D) \prod_{p|u/u_e} r_j(p; D)$. $\square$

Lemma 12 (the newform periods). *Let v be a newform of level u with Fricke eigenvalue ε , under the hypotheses of Lemma 6. Then $\text{Per}_{\mathcal{W}_u}(v(u \cdot)) = \varepsilon \text{Per}_{\mathcal{W}_u}(v)$, and*

$$\text{Per}_{\mathcal{W}_u}(v) = \begin{cases} 0 & u \text{ inert,} \\ -\sum_C |S_C|^{-1} \sum_{\ell_2 \text{ isotropic}} v(h_{\ell_2}(z_C)) & u \text{ split,} \\ (u-1) \sum_C |S_C|^{-1} v(h_r(z_C)) & u \parallel D, \end{cases}$$

where the $h_\ell(z_C)$ are taken as points of $X_0(u)$: the sums on the right are periods of v over the Heegner points of level u and discriminant $4D$ above each class, in the sense of Gross–Kohnen–Zagier. In particular $\text{Per}_{\mathcal{W}_u}(v) = 0$ whenever $\varepsilon = -1$, whatever the splitting of u .

Proof. For fixed ℓ_2 the u choices of ℓ_1 are $\gamma \mapsto \gamma \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix}$, and $\alpha \begin{pmatrix} 1 & 0 \\ -c & 1 \end{pmatrix} \alpha^{-1} = \begin{pmatrix} 1 & 0 \\ -uc & 1 \end{pmatrix} \in \Gamma_0(u)$; so $v(\alpha \gamma^{-1} z_0) =: V(\ell_2)$ depends only on ℓ_2 also for the level- u form. As points of $X_0(u)$ the $u+1$ values are $(z_0 + k)/u$ ($\ell_2 = \langle (-k, 1) \rangle$) and $-1/(uz_0)$ ($\ell_2 = \langle e_1 \rangle$), so $\sum_{\ell_2} V(\ell_2) = \sqrt{u} a_v(u) v(z_0) + \varepsilon v(z_0) = (-\varepsilon + \varepsilon) v(z_0) = 0$: for a newform the Hecke term and the Fricke term cancel. By (3), $\text{Per}_{\mathcal{W}_u}(v) = \sum_C |S_C|^{-1} \sum_{\ell_2} N(\ell_2) V_C(\ell_2)$ with N as in Lemma 7; for inert u , $N \equiv 1$ and the sum vanishes; for split u , $N = 1 - \mathbf{1}[\text{isotropic}]$; for $u \parallel D$, $N(r) = u$ and $N = 1$ otherwise.

The relation between v and $v(u \cdot)$ is the Fricke involution of $\Gamma_0(u^2)$ acting on $\mathcal{W}_u$ as the divisor pairing, since $v(-1/(u^2\tau)) = \varepsilon v(u\tau)$. Numerically the split and ramified formulas reproduce the direct orbit sums to all computed digits at $D = -8, -11, -20$ and $D = -3, -15$ ($u = 3$). For the last sentence: the Fricke involution $z \mapsto -1/(uz)$ of $\Gamma_0(u)$ maps the Heegner point of $[ua, b, c]$ to that of $[uc, -b, a]$, a bijection between the classes with $b \equiv \beta$ and $b \equiv -\beta \pmod{2u}$ that preserves stabilisers. In the split case the isotropic lines give the two root classes, which it exchanges, so the sum is $(1 + \varepsilon)$ times the sum over one of them. In the ramified case it fixes the class $b \equiv 0$, so the sum equals ε times itself. Numerically, for the even newform of level 5 with $R = 5.4362$ and $\varepsilon = -1$ (LMFDB 5.0.1.5.1) the family period computed from the orbits of $\mathcal{W}_5$ is below 10^{-9} times the sum of the absolute values of its terms at $D = -4, -11$ (split), -15 (ramified) and $-3, -8$ (inert); for the forms with $R = 4.1324$ and 6.0540 ($\varepsilon = +1$) that ratio is between 0.03 and 1 in the split and ramified cases, and $\text{Per}_{\mathcal{W}_5}(v(5 \cdot))/\text{Per}_{\mathcal{W}_5}(v) = +1$ (`newform-fricke-check.py`). $\square$

Remark 13 (the components of the family, and the inert identity explained). Let u be an odd prime with $u \nmid D$. By (3) the piece over a class C is a sum over the cosets γ of the seed at $\alpha\gamma^{-1}z_C$, weighted by $w_C(\gamma) = \mathbf{1}[\ell_1 \perp \ell_2]$. On the $u(u+1)$ ordered pairs of distinct lines, with the uniform measure, write

$$w_C = \bar{w} + \frac{u}{u-1}(n(\ell_1) + n(\ell_2)) + R_C, \quad \bar{w} = \frac{u - \chi}{u(u+1)}, \quad n(\ell) = \frac{N(\ell)}{u} - \bar{w},$$

with R_C orthogonal to every function of a single line. (i) Replacing w_C by $\bar{w}$ sums the seed over all cosets, which is Atkin's operator U_u followed by the level-one Poincaré series; since the Fourier coefficients of the seed are homogeneous, $\sum_{j \bmod u} \Psi_{Y'}^w((z+j)/u) = u^{-1} \Psi_Y^w(z)$ exactly, and the trivial part of piece u is $\bar{w} S_1^w(uY)$. It gives the level-one lines the ratio $\bar{w} u^{-1/2}$ for every splitting type. (ii) By the proofs of Lemmas 7 and 12, the values of f_0, f_1, f_2 and of the level- u newforms at the coset points depend on at most one line. So R_C contributes no level-one and no level- u line, and the marginal part carries $r_j(u; D) - \bar{w} u^{-1/2}$ and all level- u newform lines. (iii) For inert u no line is isotropic, $N \equiv 1 = u\bar{w}$, the marginal part vanishes, and $r_j(u; D) = \bar{w} u^{-1/2} = u^{-3/2}$. Arithmetically, piece u is the restriction of the $u = 1$ sum at length uY to $h' \equiv 0 \pmod{u}$; for a divisor a of $h'^2 - D$, ℓ_1 is isotropic iff $u \mid a$ and ℓ_2 iff $u \mid (h'^2 - D)/a$. The three parts are therefore computable from $\sigma_{-1}(h'^2 - D)$ and $\sigma_{-1}((h'^2 - D)/u)$, and they are tested on the data in Section 4.

Proof of Theorem 2 from the lemmas. Theorem 1 restricted to the span of f_0, f_1, f_2 gives the coefficient $\sqrt{u} \sum_k \text{Per}_{\mathcal{W}_u}(v_k) \tilde{L}(v_k)$ over an orthonormal basis v_k of that span, which is the basis-free quantity $\sqrt{u} \mathbf{P}^T \mathbf{G}^{-1} \mathbf{L}$ with $\mathbf{P}_m = \text{Per}_{\mathcal{W}_u}(f_m)$, $\mathbf{G}_{mn} = \langle f_m, f_n \rangle$, $\mathbf{L}_m = \tilde{L}_{f_m}$. Dividing by the $u = 1$ coefficient $\text{Per}_D(u_j) \tilde{L}_j / \|u_j\|^2$ and inserting the lemmas gives (2); the phase u^{-it_j} is from Theorem 1. The asymptotics are not read off termwise — $p_1 \asymp u$ is large while $L_2 = 1/u$ and $g \asymp \lambda u^{-1/2}$ are small, and the $\lambda_j(u) \sqrt{u}$ of p_0 cancels against $p_1(1/u - g)$ — but from the closed form (2) as a rational function of λ and $\sqrt{u}$: with a computer algebra system, $r = u^{-3/2}$ identically when $\chi = -1$, and the expansions in $u^{-1/2}$ stated in the theorem when $\chi = +1$ and $\chi = 0$. $\square$

4 Numerical confirmation

All data are the pieces of part IV computed to $Y = 10^7$ (`piece-divset.ts`, unrestricted object), sharp Riesz means on 4096 log-spaced points and smooth windows on 400; amplitudes and phases by joint least squares at the level-one lines with the dominant level- u^2 lines as nuisance

frequencies (Strömberg’s $\Gamma_0(9)$ eigenvalues at $u = 3$ [7], the six strongest data-driven peaks at $u = 5$). The predictions of Theorem 2 were computed twice: from (2), and directly by evaluating the Maass form at the orbit representatives of $\mathcal{W}_u$, integrating the Gram matrix over the cosets of $\Gamma_0(u^2)$, and orthonormalising; the two agree to four digits, and the direct route also checks $\langle f_0, f_0 \rangle / \|u_j\|^2 = u(u+1)$ and the Iwaniec–Luo–Sarnak entry.

Table 1: The first even line $t_1 = 13.7798$: signed ratio of its amplitude in piece u to piece 1 (sign – when the phase is shifted by π beyond $-t_1 \log u$). “pred” from (2), with the class-by-class P of Proposition 8 for $u = 2$; “obs” from the sharp grids. Errors of the observed ratios, estimated in two ways (the amplitude fitted at off-line probe frequencies, and half the difference of the fits on the two halves of the range; `ratio-noise-audit.py`, `ratio-halves-audit.py`): 0.01–0.04 for the $u = 2$ inert rows, 0.04–0.3 for the other rows at $u = 2$, 0.04–0.5 at $u = 3$ and 0.02–0.7 at $u = 5$, where the level- u^2 spectrum is dense. Every entry lies within about one error of its prediction.

u , type of u in $\mathbb{Q}(\sqrt{D})$	pred		obs	D
2, $D \equiv 0 \pmod{4}$, $D/4 \equiv 2, 3 \pmod{4}$	+0.850	+0.852, +0.865, +0.899	−4, −8, −20	
2, $D \equiv 0 \pmod{4}$, $D/4 \equiv 1 \pmod{4}$	+0.945	(piece $u = 2$ not yet fitted)	−12	
2, split ($D \equiv 1 \pmod{8}$)	−0.341	−0.353, −0.379, −0.428	−7, −15, −23	
2, inert ($D \equiv 5 \pmod{8}$)	−0.068	−0.065, −0.061, −0.071	−3, −11, −19	
3, ramified	+0.695	+0.655, +0.726	−3, −15	
3, split	−0.311	−0.285, −0.320, −0.332	−8, −11, −20	
3, inert	+0.192	+0.196, +0.341	−4, −7	
5, ramified	+0.421	+0.431	−15	
5, split	−0.076	(noise)	−4, −11	
5, inert	+0.089	(noise)	−3, −8	

Table 2: The first even newform of level 3 ($t = 5.0987$, Fricke +1) in the smooth piece $u = 3$: predicted amplitude and phase from Theorem 3 with the LMFDB coefficients, against the fit (all $\Gamma_0(9)$ even lines and four level-one lines in the fit).

D	3 in $\mathbb{Q}(\sqrt{D})$	C_{pred}	C_{obs}	ratio	$\Delta\varphi$
–3	ramified	0.3200	0.3262	1.020	–0.008
–15	ramified	0.2016	0.1946	0.965	+0.024
–8	split	0.1404	0.1426	1.016	+0.072
–11	split	0.1371	0.1370	1.000	+0.050
–20	split	0.1945	0.1931	0.993	+0.032
–4	inert	0	0.019	—	—
–7	inert	0	0.006	—	—

The case $u = 2$. The local constants are not integers ($p_1 = 4.913, 5.278, 0.321, 1.373$ for $D/4 \equiv 2, 3 \pmod{4}$, $D/4 \equiv 1 \pmod{4}$, $D \equiv 1 \pmod{8}$, $D \equiv 5 \pmod{8}$): the multiplicity depends on the class through B_C modulo 2, Proposition 8, and (2) applies with the class-by-class P . The values so obtained coincide with the direct computation and match the data (Table 1).

Composite u . For squarefree $u = u_1 u_2$ the oldform space is $\{u_j(dz) : d \mid u^2\}$, of dimension 9 for $u = 6$, and Proposition 11 gives $r_j(u; D) = \prod_{p|u} r_j(p; D)$. The direct computation at level 36 (Gram matrix over the 72 cosets, $\|f_1\|^2 / \|u_j\|^2 = 72.000$) was made before

the proposition was proved and confirms it: $r_1(6; -12) = +0.6573$ and $r_1(6; -8) = -0.2642$, against the products $r_1(2; -12)r_1(3; -12) = 0.945 \times 0.695 = 0.657$ and $r_1(2; -8)r_1(3; -8) = 0.850 \times (-0.311) = -0.264$; and the 9×9 Gram matrix is the tensor product of the 3×3 matrices at levels 4 and 9 entry by entry ($0.0781 = 0.7303 \times 0.1069$, $-0.2323 = 0.7303 \times (-0.3181)$, $-0.0955 = 0.3001 \times (-0.3181)$, ...). The data at $u = 6$ are noisy (the level-36 spectrum is dense): the five signs observed agree with the products and the magnitudes scatter by up to 0.2 ($D = -8, -20, -15, -4, -12$: observed $-0.449, -0.320, -0.295, +0.111, +0.549$ against $-0.264, -0.264, -0.237, +0.163, +0.657$).

The cycloidal lines. At $u = 3$ the pieces are dominated by the lines 3.536, 5.504, 6.647, 7.432, 8.698, which Strömberg attributes to forms of the cycloidal group Γ^3 [7]; their coefficients are not tabulated and their amplitudes in piece 3 cannot be predicted here.

The components of the piece. Remark 13 was tested on the same grids for $u = 3$ ($D = -8, -11, -20$ split, $-4, -7$ inert) and $u = 5$ ($D = -4, -11$ split, -8 inert), with the three parts computed from $\sigma_{-1}(h'^2 - D)$ and $\sigma_{-1}((h'^2 - D)/u)$ (`hecke-components.ts`); their sum reproduces the grids of part IV. For inert u the marginal part vanishes identically. At $u = 3$ the level-3 newform line 5.0987 lies entirely in the marginal part: amplitudes 0.143, 0.138, 0.191 there against 0.141, 0.136, 0.192 in the piece, and at most 0.007, at the noise, in R . The cycloidal lines 3.536 and 6.647 lie entirely in R (for example 0.2009 against 0.2010 at $D = -4$), with at most 0.008 in the marginal part. The trivial part carries t_1 with the ratio $\bar{w}u^{-1/2}$: observed $+0.100, +0.090, +0.083$ against $+0.096$ (split) and $+0.193, +0.188$ against $+0.192$ (inert). In the marginal part and in R the line t_1 is at or below the noise of these grids, which is consistent with the prediction but does not test it. At $u = 5$ the marginal part carries the even level-5 newforms 4.1324, 6.0540 and 8.2947, all with $\varepsilon = +1$ in the LMFDB, at 9 to 27 times its noise for $D = -4$ and 4 to 12 times for $D = -11$; the forms 5.4362 and 7.9865 with $\varepsilon = -1$ are absent from it, as Lemma 12 requires. R carries lines at 2.64–2.71, at 5.22–5.28 and near 10.8, none at a level-5 newform. By size the trivial part is negligible (root mean square 0.007–0.022) against the marginal part (0.08–0.17) and R (0.15–0.32): what is not uniform in u lives at the levels u and u^2 .

Precision of the subtracted means. The means $\mathbb{E}\sigma_{-1}^*$ subtracted in all grids are floating-point Euler products over 10^6 primes or more. Against a log-sum their rounding and tail errors lie between 10^{-12} and $4 \cdot 10^{-11}$. An error δ adds about $-0.111 \delta Y^{3/2}$ to $\sqrt{Y} S_u^w(Y)$, up to 0.16 at $Y = 10^7$, and polynomial detrending in $\log Y$ does not remove it; the components above were fitted with a $Y^{3/2}$ term. Refitting Table 1 with such terms moves the $u = 2$ entries by at most 0.015, the $u = 3$ entries by at most 0.04 (split $-0.315, -0.353, -0.354$; ramified $+0.618, +0.707$; inert $+0.177$) and the $u = 5$ ramified entry to $+0.445$. The amplitudes of Table 2 move by at most 2% and the phases by at most 0.02 rad. The tables keep the original fits. These systematic shifts are smaller than the statistical errors in the caption of Table 1, which are 0.04–0.5 at $u = 3$: the table shows consistency with (2) within those errors, not agreement to a few percent.

5 What is open

1. Closed forms for the class-dependent entries of Proposition 8 (at $u = 2$, and for $u^2 \mid D$) in terms of $\lambda_j(u)$ and the periods of the discriminants D/u^{2k} , via the Hecke relations. These entries can exceed 1 (Remark 9); for fixed D they enter Corollary 4 only through a constant, and for such u the decay comes from the other primes.

2. The lines of the cyclotomic-group forms, and any bound for the newform lines uniform in the newform: the level-aspect equidistribution of [4, 2], route 1b of the knowledge base.
3. The error term uniform in u (part IV, Remark GAPS), which is the conjecture of part III itself. This is not addressed here but in part V [6], which bounds the size of the pieces rather than describing their spectrum: power savings are proved for the dilations $u > H^{1/2+\varepsilon}$ (part III, dispersion over u), for $H^{0.47} < u \leq H^{0.53}$ (part V, Theorem M, via the additive energy of reciprocals and the bilinear bound of Bourgain and Garaev) and, for $D < 0$, for $u \leq H^{1/3-c}$ (part V, Theorem S, via the automorphic kernel method of Grimmer and Merikoski at level u^2 , where the off-diagonal count does not grow with u because the distances inside the family are quantised in steps of $1/(4h)$ — the same level- u^2 family whose spectrum is described here). What remains open is a neighbourhood of $u = H^{1/3}$, and part V records exactly what that band costs.

AI-assisted research: disclosure

This work was carried out with heavy use of large language models, the Anthropic models Claude Fable 5.1 and Claude Opus 5, used interactively by the author in September 2026 as research assistants. The author posed the questions, set the direction and the scope, and made every decision about what to pursue, claim and submit. The model derived the identities, wrote and ran the software behind every number, searched the literature and drafted the text, in dialogue with the author, who verified the code and the numerical claims against the repository record. The repository (<https://github.com/gewure/ulamnd>) contains the scripts, the data, and the knowledge base with the record of errors.

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